CS: 460 **Computer Vision** **Spring, 2026**

Project5 (100 points) **Due 3/3/2026**

Please submit your solutions to canvas.

https://towardsdatascience.com/understanding-ssd-multibox-real-time-object-detection-in-deep-learning-495ef744fab/

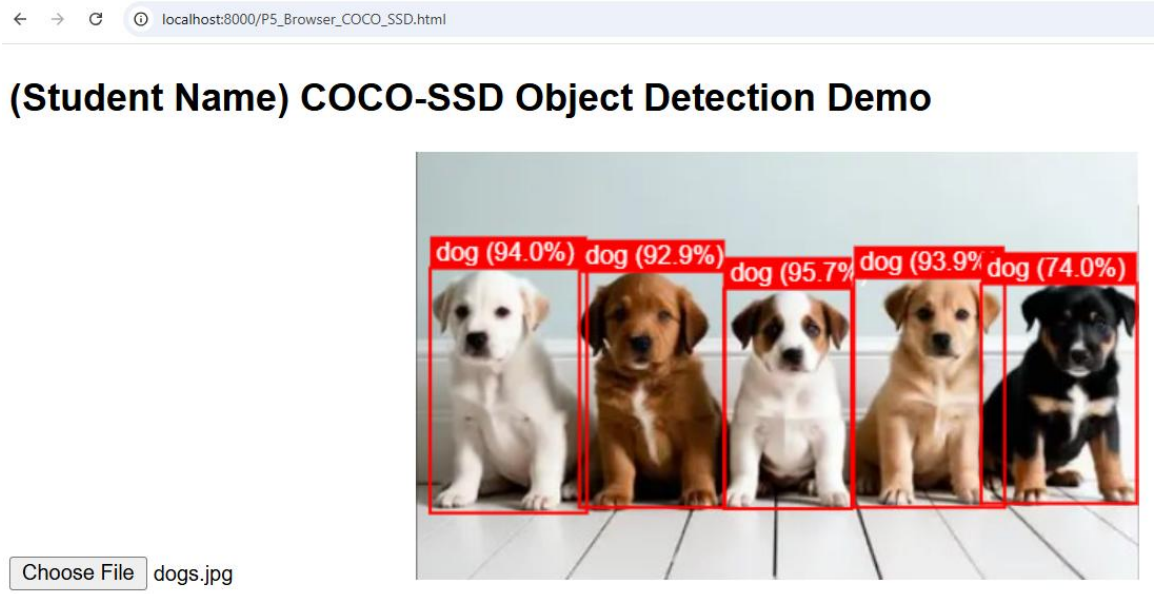
Assignment Objects:

- Understand how a pretrained convolutional neural network (COCO-SSD) performs object detection.
- Learn how to visualize detection results with bounding boxes and confidence scores.

Problem 1 (Static Image to File Browser Detection)

Use the file files `P5_starter_COCO_SSD.html` and `P5_starter_COCO_SSD.js` to make one app that allows you to read in a file and make a classification of the image file read in. Initially the prediction will show up in the console window. Modify the program so the prediction also shows up in the HTML canvas with bounding boxes and probabilities.

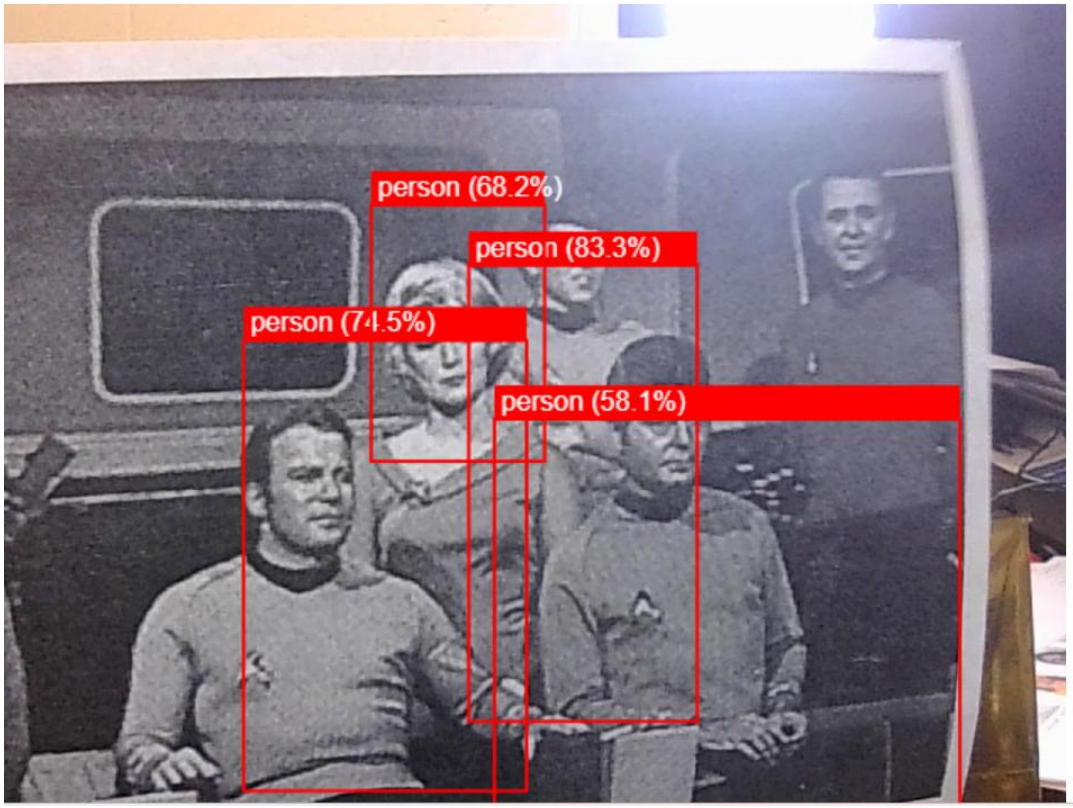
Your app should look like this:



Problem 2 (Real-Time Webcam Detection)

Modify your solution to read from a webcam instead of a static file. The detection should run continuously on live video.

The tasks are:

1. Replace the <input> and <img> with a <video> element.
2. Access the webcam using navigator.mediaDevices.getUserMedia({ video: true }).
3. Run model.detect(video) on each frame.
4. Draw bounding boxes and labels on a <canvas> overlay.
5. Optimize performance using requestAnimationFrame.

Student Name) COCO-SSD Webcam Demo